

MINGEUN CHOI

Ph.D. Student

George W. Woodruff School of Mechanical Engineering

Georgia Institute of Technology

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🌐 [linkedin.com/in/mingeunchoi](https://www.linkedin.com/in/mingeunchoi)

EDUCATION

Georgia Institute of Technology

Atlanta, GA

Ph.D. in Mechanical Engineering (Advisor: Dr. Satish Kumar)

Jan. 2023 – Dec. 2027 (expected)

Seoul National University

Seoul, Republic of Korea

M.S. in Mechanical Engineering (Advisor: Dr. Seung Jin Song)

Mar. 2017 – Feb. 2019

Korea University

Seoul, Republic of Korea

B.S. in Mechanical Engineering

Mar. 2013 – Feb. 2017

APPOINTMENTS

Republic of Korea Air Force Academy

Cheongju, Republic of Korea

Assistant Professor (Rank: First Lieutenant)

Jun. 2020 – May 2022

Instructor (Rank: Second Lieutenant)

Jun. 2019 – May 2020

RESEARCH INTERESTS

My research focuses on electrothermal modeling and thin-film thermal-property characterization in heterogeneously integrated electronics. I develop (1) an automated electrothermal modeling framework to co-optimize vertical power delivery architecture and substrate-embedded microfluidic cooling for thermally sustainable, high-power-density operation; (2) a machine-learning-accelerated electrothermal modeling framework to predict spatiotemporal temperature fields from individual FinFET devices to arrays for fast, accurate, self-heating-aware evaluation; and (3) a machine-learning-enabled frequency-domain thermoreflectance inversion model to estimate the thermal properties of thin films for high-throughput, low-uncertainty characterization. Together, I aim to build a self-consistent multiscale workflow linking the device and package levels with high-fidelity thermal-property inputs.

PUBLICATIONS

Refereed Journal Articles

- [8] **M. Choi**, S. Krishnakumar, Y. Popryho, R. R. Khorasani, M. Swaminathan, I. Partin-Vaisband, and S. Kumar, "Self-Consistent Electrothermal Modeling of Distributed Vertical Power Delivery Architecture with Substrate-Embedded Microfluidic Cooling," *IEEE Trans. Compon., Packag., Manuf. Technol.*, early access, Nov. 24, 2025, doi: [10.1109/TCPMT.2025.3636391](https://doi.org/10.1109/TCPMT.2025.3636391).
- [7] D. Go, V. Ashita, A. Wang, **M. Choi**, H. S. Song, S. Yun, D. Pal, P.-C. Lee, J. Dutta, J. Ryu, S. Kumar, C. V. Thompson, M. S. Bakir, A. C. Kummel, "Chiplet Encapsulation and Planarization with 100 μm Deep Gaps using AlN Powder Spray + Sputter Hybrid Deposition," *Adv. Eng. Mater.*, vol. 28, no. 6, pp. 1438-1656, Mar. 2026, doi: [10.1002/adem.202502821](https://doi.org/10.1002/adem.202502821).
- [6] S. Yun, D. Go, **M. Choi**, R. Kondakindi, P.-C. Lee, P. R. Bandaru, S. Kumar, and A. C. Kummel, "High Speed, High Thermal-Conductivity of Aluminum Nitride Deposited by DC Reactive Sputtering at Low Temperature in the Transition Regime," *Ceram. Int.*, vol. 52, no. 4, pp. 4777-4786, Feb. 2026, doi: [10.1016/j.ceramint.2025.12.249](https://doi.org/10.1016/j.ceramint.2025.12.249).
- [5] P.-C. Lee, **M. Choi**, D. Contreras Mora, K. Wang, S. Yun, D. Go, J. Dutta, D. Pal, S. Kumar, and A. C. Kummel, "High-Speed AlN Film Deposition via Low-Pressure Bipolar High Power Impulsed

- Magnetron Sputter for Enhanced Thermal Conductivity," *Thin Solid Films*, vol. 832, p. 140821, Dec. 2025, doi: [10.1016/j.tsf.2025.140821](https://doi.org/10.1016/j.tsf.2025.140821).
- [4] **M. Choi**, D. Vaca, and S. Kumar, "Machine Learning-Enabled Fast and Accurate Inversion of Thermal Properties at the Micro- and Nanoscale via Optical Metrology," *Annu. Rev. Heat Transfer*, vol. 28, pp. 277-337, Oct. 2025, doi: [10.1615/AnnualRevHeatTransfer.2025059842](https://doi.org/10.1615/AnnualRevHeatTransfer.2025059842).
- [3] **M. Choi**, S. Krishnakumar, R. R. Khorasani, M. Swaminathan, I. Partin-Vaisband, and S. Kumar, "Substrate-Embedded Microfluidic Cooling of Distributed Vertical Power Delivery Architectures for High-Performance Computing Processors," *IEEE Trans. Compon., Packag., Manuf. Technol.*, vol. 15, no. 9, pp. 1912-1920, Sep. 2025, doi: [10.1109/TCPMT.2025.3544105](https://doi.org/10.1109/TCPMT.2025.3544105).
- [2] P.-C. Lee, A. J. McLeod, **M. Choi**, D. Vaca, D. Contreras Mora, K. Wang, S. Yun, J. Dutta, D. Pal, S. Kumar, and A. C. Kummel, "Achieving a High Thermally Conductive One Micron AlN Deposition by High Power Impulse Magnetron Sputtering Plus Kick," *ACS Appl. Mater. Interfaces*, vol. 16, no. 20, pp. 26664-26673, May 2024, doi: [10.1021/acsami.4c00993](https://doi.org/10.1021/acsami.4c00993).
- [1] **M. Choi**, B. Goo, G. Cho, and S. J. Song, "Swirl Enhancement Effect on Turbine Rim Seal Performance," *ASME J. Turbomach.*, vol. 146, no. 5, p. 051002, May 2024, doi: [10.1115/1.4064230](https://doi.org/10.1115/1.4064230).

Manuscripts Under Review

- [3] **M. Choi**, R. Dutta, P. Saha, M. Singh, S. Mukhopadhyay, S. Datta, and S. Kumar, "Deep Operator Network-Accelerated Spatiotemporal Temperature Field Assessment of Fin Field-Effect Transistor Arrays," submitted for publication.
- [2] P.-C. Lee, **M. Choi**, D. Contreras Mora, A. Ross, S. Yun, D. Go, J. Dutta, X. Wang, M. Clark, L. Chen, R. Kanjolia, M. Moinpour, H. Chandra, S. Kumar, and A. C. Kummel, "Ballistic Thermal Transport in Vertically Aligned h-BN," submitted for publication.
- [1] S. Krishnakumar, Y. Popryho, **M. Choi**, R. R. Khorasani, M. Swaminathan, S. Kumar, and I. Partin-Vaisband, "A Comprehensive Design Framework for Vertical Power Delivery in High-Performance Computing," submitted for publication.

Refereed Conference Proceedings

- [12] **Best Paper** **M. Choi**, M. Herndon, K. Uechi, N. Akel, and S. Kumar, "DeepONet-Enabled Fast and Scenario-Aware Spatial Temperature Prediction of a Vertical Power Delivery Module for Hyperscale Data Centers," in *Proc. 25th IEEE Intersoc. Conf. Thermal Thermomech. Phenomena Electron. Syst. (ITherm)*, Orlando, FL, USA, May. 2026, to be published.
- [11] P.-C. Lee, D. Contreras Mora, **M. Choi**, S. Yun, D. Pal, A. Ross, M. Clark, L. Chen, H. Chandra, R. Kanjolia, M. Moinpour, S. Kumar, and A. Kummel, "Vertical aligned Hexagonal Boron Nitride with Ballistic Thermal Conductivity," in *Proc. IEEE Int. Symp. VLSI Technol. Syst. Appl. (VLSI-TSA)*, Hsinchu, Taiwan, Apr. 2026, pp. 1-2, doi: [10.1109/VLSITSA69131.2026.11527934](https://doi.org/10.1109/VLSITSA69131.2026.11527934).
- [10] R. Dutta, M. Shon, S. Lim, P. Saha, **M. Choi**, M. P. Singh, S. Kumar, S. Mukhopadhyay, S. K. Lim, S. Yu, and S. Datta, "Self-Heating in GAA Nanosheet FETs with integrated Back Side Power Delivery Network: Device to Circuit Electro-Thermal Analysis and Implications," in *Proc. IEEE Int. Reliab. Phys. Symp. (IRPS)*, Tucson, AZ, USA, Mar. 2026, doi: [10.1109/IRPS61424.2026.11499271](https://doi.org/10.1109/IRPS61424.2026.11499271).
- [9] **Best Paper** **News Release** **News Release** **M. Choi**, S. Krishnakumar, Y. Popryho, R. R. Khorasani, M. Swaminathan, I. Partin-Vaisband, and S. Kumar, "Automated Electro-Thermal Modeling Framework of Distributed Vertical Power Delivery Architectures with Substrate-Embedded Microfluidic Cooling," in *Proc. 24th IEEE Intersoc. Conf. Thermal Thermomech. Phenomena Electron. Syst. (ITherm)*, Dallas, TX, USA, May 2025, doi: [10.1109/ITherm55376.2025.11235621](https://doi.org/10.1109/ITherm55376.2025.11235621).
- [8] **News Release** **M. Choi**, R. Dutta, P. Saha, M. P. Singh, S. Mukhopadhyay, S. Datta, and S. Kumar, "Fast Prediction of Spatio-Temporal Temperature Profiles in FinFET Arrays via Numerical and Machine-Learning Approaches," in *Proc. IEEE Int. Electron Devices Meeting (IEDM)*, San Francisco, CA, USA, Dec. 2024, doi: [10.1109/IEDM50854.2024.10873584](https://doi.org/10.1109/IEDM50854.2024.10873584).

- [7] Y. Im, J. Kim, **M. Choi**, M. Bouzidi, X. Li, J. W. Kim, A. M. Muslu, S. Kumar, M. Swaminathan, S. K. Sitaraman, and Y. Joshi, "Parametric Thermal Design for Heterogeneously Integrated High-Power Packages," in *Proc. ASME 2024 Int. Tech. Conf. Exhib. Packaging Integr. Electron. Photon. Microsystems (InterPACK)*, San Jose, CA, USA, Oct. 2024, doi: [10.1115/IPACK2024-141216](https://doi.org/10.1115/IPACK2024-141216).
- [6] P.-C. Lee, A. J. McLeod, **M. Choi**, D. Vaca, S. Kumar, and A. C. Kummel, "Thermal Conductivity Study of One Micron AlN Deposition by Bipolar High Power Impulse Magnetron Sputtering," in *Proc. IEEE Int. Interconnect Technol. Conf. (IITC)*, San Jose, CA, USA, Jun. 2024, doi: [10.1109/IITC61274.2024.10732369](https://doi.org/10.1109/IITC61274.2024.10732369).
- [5] S. Krishnakumar, **M. Choi**, R. R. Khorasani, R. Sharma, M. Swaminathan, S. Kumar, and I. Partin-Vaisband, "Vertical Power Delivery for High Performance Computing Systems with Buck-Derived Regulators," in *Proc. IEEE 74th Electron. Compon. Technol. Conf. (ECTC)*, Denver, CO, USA, May 2024, doi: [10.1109/ECTC51529.2024.00364](https://doi.org/10.1109/ECTC51529.2024.00364).
- [4] **M. Choi**, S. Krishnakumar, R. R. Khorasani, I. Partin-Vaisband, R. Sharma, M. Swaminathan, and S. Kumar, "Thermal Analysis of High Current Vertical Power Delivery Network with Embedded Microchannel Cooling," in *Proc. 23rd IEEE Intersoc. Conf. Thermal Thermomech. Phenomena Electron. Syst. (ITherm)*, Aurora, CO, USA, May 2024, doi: [10.1109/ITherm53375.2024.10709419](https://doi.org/10.1109/ITherm53375.2024.10709419).
- [3] S. Krishnakumar, **M. Choi**, R. R. Khorasani, R. Sharma, M. Swaminathan, S. Kumar, and I. Partin-Vaisband, "Design Considerations for DC-DC Voltage Regulators in Distributed Vertical Power Delivery Systems," in *Proc. IEEE Int. Symp. Circuits Syst. (ISCAS)*, Singapore, May 2024, doi: [10.1109/IS-CAS58744.2024.10558456](https://doi.org/10.1109/IS-CAS58744.2024.10558456).
- [2] P.-C. Lee, A. J. McLeod, **M. Choi**, D. Vaca, D. Contreras Mora, S. Kumar, and A. C. Kummel, "High Thermally Conductive, High-Speed Deposition of AlN by Bipolar High Power Impulse Magnetron Sputtering," in *Proc. IEEE Int. Symp. VLSI Technol. Syst. Appl. (VLSI-TSA)*, Hsinchu, Taiwan, Apr. 2024, doi: [10.1109/VLSITSA60681.2024.10546374](https://doi.org/10.1109/VLSITSA60681.2024.10546374).
- [1] **M. Choi**, B. Goo, G. Cho, and S. J. Song, "Swirl Enhancement Effect on Turbine Rim Seal Performance," in *Proc. ASME Turbo Expo 2023: Turbomach. Tech. Conf. Expo*, Boston, MA, USA, Jun. 2023, doi: [10.1115/GT2023-101389](https://doi.org/10.1115/GT2023-101389).

Conference Presentations

- [11] M. Singh, **M. Choi**, L. Patra, R. Dutta, P. Saha, S. Mukhopadhyay, S. Datta, and S. Kumar, "Machine Learning Enabled Fast and Accurate Thermal Model of Nanoscale Transistors and Circuits," presented at the *Gov. Microcircuit Appl. Critical Technol. Conf. (GOMACTech) 2026*, New Orleans, LA, USA, Mar. 9–12, 2026.
- [10] P.-C. Lee, D. Contreras Mora, **M. Choi**, S. Yun, D. Pal, A. Ross, M. Clark, L. Chen, R. Kanjolia, M. Moinpour, S. Kumar, and A. C. Kummel, "Vertical Aligned Hexagonal Boron Nitride with High Interfacial Thermal Conductivity," presented at the *56th IEEE Semiconductor Interface Specialists Conf. (SISC)*, San Diego, CA, USA, Dec. 10–13, 2025.
- [9] D. Contreras Mora, P.-C. Lee, **M. Choi**, D. Go, S. Yun, M. Moinpour, L. Chen, M. Clark, R. Kanjolia, S. Kumar, and A. C. Kummel, "Thermal Boundary Conductance of Ultrathin Aluminum Nitride Interfaces for Advanced Packaging," presented at the *56th IEEE Semiconductor Interface Specialists Conf. (SISC)*, San Diego, CA, USA, Dec. 10–13, 2025.
- [8] D. Contreras Mora, P.-C. Lee, **M. Choi**, S. Kumar, and A. C. Kummel, "Low Pressure HiPIMS + Kick AlN Deposition with Ultra High Growth Rate," presented at the *Gov. Microcircuit Appl. Critical Technol. Conf. (GOMACTech) 2025*, Pasadena, CA, USA, Mar. 17–20, 2025.
- [7] P.-C. Lee, D. Contreras Mora, A. J. McLeod, **M. Choi**, D. Vaca, S. Kumar, and A. C. Kummel, "Enhancing AlN Crystallographic Properties and Thermal Conductivity Using HiPIMS + Kick Deposition," presented at the *Gov. Microcircuit Appl. Critical Technol. Conf. (GOMACTech) 2024*, Charleston, SC, USA, Mar. 18–21, 2024.
- [6] P.-C. Lee, D. Contreras Mora, A. J. McLeod, **M. Choi**, D. Vaca, S. Kumar, and A. C. Kummel, "Achieving

a High Thermal Conductive One Micron AlN Deposition by High Power Impulse Magnetron Sputtering Plus Kick," presented at the *54th IEEE Semiconductor Interface Specialists Conf. (SISC)*, San Diego, CA, USA, Dec. 13–16, 2023.

- [5] J. Pak, S. Oh, and **M. Choi**, "Computational Modeling for Investigating Flow Instabilities in Gas Turbine Rim Seals," presented at the *KSPE Spring Conf. 2020*, Online via Zoom, Jul. 16–17, 2020.
- [4] **M. Choi**, B. Goo, and S. J. Song, "Design of a Low-Speed Single-Stage Turbine Facility for Ingress Measurement," presented at the *IGTC 2019 Tokyo*, Tokyo, Japan, Nov. 17–22, 2019.
- [3] **M. Choi**, B. Goo, and S. J. Song, "Sealing Performance Measurement of Single Radial-Clearance Rim Seal," presented at the *KFMA Annu. Meeting 2018*, Yeosu, Republic of Korea, Nov. 28–30, 2018.
- [2] **M. Choi**, H. R. Kim, S. Kim, G. Cho, and S. J. Song, "Measurement of Double-Clearance Rim Seal Performance to Prevent Hot Gas Ingestion in Large-Scale Gas Turbines," presented at the *KFMA Annu. Meeting 2018*, Jeju, Republic of Korea, Jul. 4–6, 2018.
- [1] **Best Paper News Release** **M. Choi**, G. Cho, and S. J. Song, "Rebuilding of 1-Stage Axial Turbine Rim Seal Test Facility in Seoul National University," presented at the *KFMA Annu. Meeting 2018*, Jeju, Republic of Korea, Jul. 4–6, 2018.

PATENTS

- [3] A. C. Kummel, P.-C. Lee, D. Contreras Mora, D. Go, S. Kumar, and **M. Choi**, "Aluminum Nitride Thin Film Deposition Using Sputtering," U.S. Patent Application No. 19/366,019, filed on Oct. 22, 2025, published on Apr. 23, 2026, [link: patents.google.com/patent/US20260110074A1](https://patents.google.com/patent/US20260110074A1).
- [2] **M. Choi** and S. J. Song, "Gas Turbine Engine with Improved Flow Stability and Cooling Efficiency," Republic of Korea Patent 10-2206447, filed on Oct. 8, 2020, issued on Jan. 18, 2021, [doi: 10.8080/1020200130286](https://doi.org/10.8080/1020200130286).
- [1] **M. Choi**, S. J. Song, and B. Goo, "Gas Turbine Engine with Improved Sealing and Operability," Republic of Korea Patent 10-2202368, filed on Sep. 16, 2020, issued on Jan. 7, 2021, [doi: 10.8080/1020200119006](https://doi.org/10.8080/1020200119006).

RESEARCH EXPERIENCE

Georgia Institute of Technology

Micro Nano Devices and Systems Laboratory

Graduate Research Assistant

Jan. 2023 – Present

- Supported by the Center for Heterogeneous Integration of Micro Electronic Systems (CHIMES), led by the Semiconductor Research Corporation (SRC) and sponsored by the Defense Advanced Research Projects Agency (DARPA) to develop advanced thermal management solutions for heterogeneously integrated electronics.
- Enabled distributed vertical power delivery (VPD) architectures with substrate-embedded microfluidic cooling in 3D-stacked processors that sustain current densities exceeding 2 A/mm² at up to 80% efficiency.
- Developed high-thermal-conductivity aluminum nitride (AlN) thin films as heat spreaders using low-pressure high-power impulse magnetron sputtering (HiPIMS) with "plus kick voltage" to achieve deposition rates above 25 nm/min at temperatures below 200°C.

Graduate Research Assistant

Aug. 2023 – Present

- Sponsored by the Thermal Modeling of Nanoscale Transistors (Thermonat) program operated by Defense Advanced Research Projects Agency (DARPA) to develop innovative multi-scale thermal management approaches for nanoscale transistors as part of their Microsystems Exploration (μE) initiative.
- Developed a machine learning-accelerated multi-scale modeling framework that predicts spatio-temporal temperature profiles in 14 nm fin field-effect transistor (FinFET) arrays nearly 2300 times faster than conventional finite-volume method (FVM).
- Developed a device-to-circuit electrothermal framework for gate-all-around nanosheet field-effect transistors (GAA NSFETs) with backside power delivery networks (BSPDN) by calibrating self-heating effect (SHE)-aware BSIM-CMG models to predict over 10% frequency degradation.

Seoul National University

Graduate Student Researcher

Turbomachinery Laboratory

Mar. 2017 – Feb. 2019

- Sponsored by Doosan Enerbility to construct and validate an engine-representative gas turbine test rig aimed at reducing hot gas ingestion and to investigate the impact of swirling sealing flows on rim sealing performance.
- Constructed a scaled test rig of the first high-pressure turbine (HPT) stage in H^+ gas turbines, and demonstrated that reducing rim seal clearance decreases hot gas ingestion by significant margins.
- Developed a patented rotor disk attachment “swirler” that enhances rim sealing performance up to 23 % by increasing swirl in the wheel-space, potentially improving overall efficiency and power of the gas turbine can be increased by up to 0.02 pp and 0.18 %.

TEACHING EXPERIENCE

Georgia Institute of Technology

Teaching Practicum

George W. Woodruff School of Mechanical Engineering

Jan. 2024 – May 2024

- Heat Transfer (Spring 2024)

ME3345

Republic of Korea Air Force Academy

Instructor – Assistant Professor

Department of Mechanical Engineering

Jun. 2019 – May 2022

- Fluid Mechanics (Spring 2020, Spring 2022) A0532
- Applied Fluid Mechanics (Spring 2021) A1154
- Thermodynamics (Fall 2022) A0438
- Heat Transfer (Fall 2019, Fall 2020, Fall 2021) A0439
- Energy Systems Engineering (Fall 2022) A1692
- Engineering Mechanics: Statics and Dynamics (Fall 2019, Fall 2020, Spring 2022) A0066
- Mechanical Component Design and Drawing (Fall 2019, Fall 2020) A0151
- Manufacturing Processes and Practice (Spring 2021) A0138
- Mechanical Engineering Laboratory (Spring 2021, Spring 2022) A0145
- Practical Research on Military Task (Fall 2020, Fall 2021) A1610

Seoul National University

Teaching Assistant

Department of Mechanical Engineering

Mar. 2018 – Dec. 2018

- Mechanical System Design Project 2 (Fall 2018) M2794.004300
- Mechanical System Design Project 1 (Spring 2018) M2794.004200
- Fluid Mechanics (Spring 2018) M2794.001300

TECHNICAL SKILLS

- **Engineering and Simulation Software:** ANSYS Icepak (including PyAEDT); ANSYS Fluent (including PyFluent); ANSYS CFX; Dassault Systèmes SolidWorks; Microsoft Visual Studio Code.
- **Programming and Scripting:** Python; MathWorks MATLAB; Julia; National Instruments LabVIEW.
- **Experimental Metrology (including Relevant Instrument):** Frequency-domain thermoreflectance (FDTR); thermoreflectance thermal imaging (TTI); thermofluidic measurement; machining; manufacturing; commissioning; calibration; uncertainty analysis.

HONORS AND AWARDS

- [13] **Professor Avram Bar-Cohen Best Paper Award** at Intersoc. Conf. Thermal Thermomech. Phenomena Electron. Syst. (ITherm) Conference, Institute of Electrical and Electronics Engineers (IEEE), 2026.
- [12] **News Release Roadmap Paper Topping Fellowship (Paul R. Yopp Memorial Fellowship)**, Georgia Institute of Technology, 2026.

- [11] **News Release** **Most First Author Journal Papers Topping Fellowship (Beth & Ford Flowers Fellowship)**, Georgia Institute of Technology, 2025.
- [10] **Best Paper Runner-Up Award** at Intersoc. Conf. Thermal Thermomech. Phenomena Electron. Syst. (ITherm) Conference, Institute of Electrical and Electronics Engineers (IEEE), 2025.
- [9] **News Release** **Excellence in Service and Leadership Award**, Georgia Institute of Technology, 2025.
- [8] **Community Builder/Leader Topping Fellowship (Beth & Ford Flowers Fellowship)**, Georgia Institute of Technology, 2025.
- [7] **Strong Gap Funding Topping Fellowship**, Georgia Institute of Technology, 2024.
- [6] **Best Presentation Award** at KFMA Annu. Meeting Conference, Korean Society for Fluid Machinery, 2018.
- [5] **Merit-Based Scholarship**, Seoul National University, 2018.
- [4] **Brain Korea 21 Plus Fellowship**, [National Research Foundation of Korea](#), 2017.
- [3] **National Science and Engineering Scholarship**, [Korea Student Aid Foundation](#), 2015 – 2016.
- [2] **Semester High Honors**, Korea University, 2014 – 2016.
- [1] **Academic Excellence Scholarship for Freshman**, Korea University, 2013.

TEACHING CERTIFICATIONS

- Tech to Teaching**, Georgia Institute of Technology Mar. 2025
- Completed the capstone teaching practicum for Heat Transfer (ME 3345) by designing and delivering four instructional sessions, earning over 90 % positive student feedback.
- CIRTL Associate**, Center for the Integration of Research, Teaching, and Learning (CIRTL) Mar. 2025
- Completed foundational training in evidence-based teaching practices and inclusive-pedagogy frameworks, applying these methods to improve classroom equity and student engagement.

LEADERSHIP AND SERVICE

- Director of Communications**, [Georgia Tech Nu Chapter of Pi Tau Sigma](#) Jan. 2025 - Dec. 2025
- Launched a monthly newsletter and renovated the chapter website, driving a 100 % increase in new-initiate turnout in Spring 2025 and boosting alumni engagement by 30 %.
- External Vice President**, [Mechanical Engineering Graduate Association \(MEGA\)](#) May 2024 - Apr. 2025
- Coordinated eight professional networking events with Dow Inc., Sandia National Laboratories (SNL), Exponent, and Kratos Defense, increasing event offerings by 50 % and graduate-student attendance by 40 % over the academic year.

PROFESSIONAL AFFILIATIONS AND SERVICE

- Peer Reviewer**, IEEE Intersociety Conference on Thermal and Thermomechanical Phenomena in Electronic Systems (ITherm) 2026
- Member**, IEEE Electron Devices Society (EDS) 2026 – Present
- Member**, IEEE Electronics Packaging Society (EPS) 2026 – Present
- Graduate Student Member**, Institute of Electrical and Electronics Engineers (IEEE) 2024 – Present

REFERENCES

Satish Kumar

Frank H. Neely Professor
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Distinguished Professor
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Seung Jin Song

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